

ANALYSTLAB AFRICA

Experience Lab Internship Programme
Data Analytics Track

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Week 5 Project Summary Report

Exploratory Analysis, KPI Development and Business Insights

HealthConnect Clinic Experience Lab

Programme	AnalystLab Africa Experience Lab
Week	Week 5
Project	HealthConnect Clinic Experience Lab
Track	Data Analytics
Tools Used	Python (Pandas, Matplotlib) SQL Power BI
Document Type	Week 5 Analytics Report — EDA, KPIs, Business Insights

Part 1 — Week 4 Foundation Review

This section briefly summarises the Week 4 foundation before moving into the Week 5 implementation. It is not a repetition of the full Week 4 submission.

Item	Summary
Problem Defined	HealthConnect Clinic has a 48.46% appointment no-show rate. The Data Analytics track will use data to identify what drives no-shows and produce insights that help reduce missed appointments.
Week 4 Approach	Phase 1: Data preparation. Phase 2: EDA and KPI calculations. Phase 3: Visualisations and findings. Phase 4: Power BI dashboard.
Resources Used	HealthConnect_Appointment_Data.csv (5,000 records, 18 columns) and HealthConnect_Data_Dictionary.xlsx.
Key Assumptions	Dataset accurately represents appointment patterns. appointment_outcome is correctly recorded. Cancelled appointments treated separately from No-Shows.
Key Limitations	Dataset is fictional — findings are illustrative only. No-show reasons are unknown. 90 records were missing distance values.
Week 5 Plan	Complete data preparation, run full EDA, calculate all 5 KPIs, produce visualisations, and document business insights.

Part 2 — Data Preparation

Before any analysis could begin, the dataset was cleaned and prepared. The original dataset was loaded into a read-only variable (`df_raw`) and all cleaning was applied to a separate working copy (`df`). The original file was never modified.

2.1 Actions Taken

Action	Method	Status
Loaded original dataset into <code>df_raw</code> — kept as read-only reference	<code>pd.read_csv()</code>	Done
Created working copy <code>df</code> for all cleaning and changes	<code>df_raw.copy()</code>	Done
Converted <code>booking_date</code> from text to datetime format	<code>pd.to_datetime()</code>	Done
Converted <code>appointment_date</code> from text to datetime format	<code>pd.to_datetime()</code>	Done
Filled 90 missing distance values with column mean (10.11 km)	<code>.fillna(mean)</code>	Done
Left <code>reminder_channel</code> nulls as-is — structurally valid where <code>reminder_sent = No</code>	Verified only	Done
Left <code>waiting_time_minutes</code> nulls as-is — expected for no-show appointments	Noted only	Done
Confirmed zero duplicate rows	<code>.duplicated()</code>	Done
Confirmed all categorical values are consistent — no unexpected labels	<code>.unique()</code>	Done
Created <code>lead_time_band</code> column (0–7, 8–14, 15–30, 31–60 days)	<code>pd.cut()</code>	Done
Created <code>patient_risk_tier</code> column (Low, Medium, High Risk)	<code>.apply()</code>	Done

2.2 Derived Columns Created

Two new columns were created to support the analysis. These are derived from existing variables and make the findings more actionable.

Column Name	Based On	Values	Purpose
lead_time_band	<code>booking_lead_days</code> grouped into four ranges	0–7 8–14 15–30 31–60 days	Allows no-show rate to be compared across booking window sizes
patient_risk_tier	<code>previous_no_shows</code> classified into three tiers	Low Medium High Risk	Makes patient risk level visible and actionable for clinic staff

Data preparation is complete. The working dataset is clean, correctly typed, and ready for full analysis. The original dataset remains unchanged.

Part 3 — Exploratory Data Analysis

The EDA investigated appointment attendance and no-show patterns across 12 dimensions. For each dimension, the no-show rate was calculated and a chart was produced. All charts are saved as PNG files in the notebook folder.

3.1 Overall Outcome Distribution

Outcome	Count	Percentage	Operational Impact
No-Show	2,423	48.46%	Nearly 1 in 2 appointments wasted
Attended	2,314	46.28%	Completed consultations
Cancelled	263	5.26%	Treated separately — patient gave notice
Total	5,000	100%	

Key observation: 48.46% of all appointments result in a No-Show — nearly 1 in every 2 appointments is missed. This is the core problem the analysis aims to address.

3.2 No-Show Rate by Appointment Type

Appointment Type	No-Show Rate	vs Overall Average
Follow-up	51.23%	▲ Above average
Diagnostic Test	49.75%	▲ Above average
Specialist Consultation	47.44%	▼ Below average
General Consultation	46.64%	▼ Below average

Observation: Follow-up appointments have the highest no-show rate at 51.23%, sitting above the overall average of 48.46%. General Consultations have the lowest rate at 46.64%. Patients may assign less urgency to follow-ups, making them more likely to skip.

3.3 No-Show Rate by Day of the Week

Day	No-Show Rate	Pattern
Sunday	50.47%	Highest
Wednesday	49.66%	Above average
Monday	49.64%	Above average
Thursday	48.98%	Near average
Morning (avg)	48.46%	Overall average
Saturday	46.82%	Below average
Friday	46.57%	Lowest

Observation: Sunday (50.47%) and Monday (49.64%) have the highest no-show rates. Friday (46.57%) and Saturday (46.82%) are the lowest. Weekend-start and week-start appointments carry the greatest risk.

3.4 No-Show Rate by Time Slot

Time Slot	No-Show Rate	Note
Evening	49.78%	Highest
Afternoon	48.38%	Near average
Morning	48.14%	Lowest

Observation: Differences across time slots are small. Evening appointments carry a slightly higher risk (49.78%), but time of day is a weaker predictor compared to lead time and prior no-show history.

3.5 No-Show Rate by Age Group

Age Group	No-Show Rate	Note
55–64	50.75%	Highest
25–34	50.70%	Above average
18–24	50.18%	Above average
35–44	48.35%	Near average
45–54	48.05%	Near average
65+	45.12%	Lowest — most reliable group

Observation: Older patients (65+) are the most reliable attenders with a 45.12% no-show rate. The 55–64 and 25–34 groups have the highest rates. Younger and middle-aged patients may benefit most from targeted engagement strategies.

3.6 No-Show Rate by Booking Lead Time Band

This is one of the most important findings in the entire dataset. The no-show rate more than doubles as the booking window grows longer.

Lead Time Band	No-Show Rate	Count	Interpretation
0–7 days (same week)	27.81%	n = 678	Lowest risk — recent booking keeps it top of mind
8–14 days	33.55%	n = 659	Moderate — still relatively recent
15–30 days	43.21%	n = 1,629	Rising — approaching the overall average
31–60 days	60.49%	n = 2,034	Highest risk — more than double the same-week rate

Critical finding: Appointments booked 31–60 days in advance have a no-show rate of 60.49% — more than double that of same-week bookings (27.81%). This is the most actionable finding in the analysis.

3.7 No-Show Rate by Previous No-Show History

A patient's own history of missing appointments is the strongest behavioural predictor of future no-shows in the dataset.

Prior No-Shows	No-Show Rate	Patient Count	Risk Level
0 (none)	43.51%	2,921	Low Risk
1	53.49%	1,548	Medium Risk
2	59.36%	438	High Risk
3	67.95%	78	High Risk
4	66.67%	12	High Risk
5	100.00%	3	Extreme Risk

3.8 Reminder Effectiveness Analysis

Group	No-Show Rate	Count	vs No Reminder
No reminder sent	51.39%	1,366	Baseline
Reminder sent (all channels)	47.36%	3,634	−4.03 pp
SMS reminder	45.75%	2,000	−5.64 pp
Email reminder	48.41%	533	−2.98 pp
WhatsApp reminder	49.77%	1,101	−1.62 pp

Observation: SMS is by far the most effective reminder channel, reducing no-shows by 5.64 percentage points versus no reminder. WhatsApp produces almost no improvement, barely outperforming the no-reminder baseline.

3.9 Distance to Clinic Analysis

Outcome	Avg Distance to Clinic	Difference
Attended	9.67 km	—
Cancelled	10.11 km	+0.44 km
No-Show	10.53 km	+0.86 km

Observation: No-show patients live on average 0.86 km further from the clinic than attended patients. The relationship exists but is small — distance alone is not a strong predictor. It likely amplifies other risk factors.

3.10 Cancellations — Separate from No-Shows

Cancellations (5.26% of all appointments) were analysed separately from no-shows throughout this report. A cancellation means the patient notified the clinic in advance — this is fundamentally different from a no-show where no notice is given. Cancellations were excluded from all no-show rate calculations to avoid distorting the results.

Part 4 — KPI Development

All 5 KPIs proposed in Week 4 were calculated. Each KPI is defined, calculated, linked to a business question, and interpreted below.

KPI 1 — Overall Appointment No-Show Rate

Detail	Description
Linked to	BQ1 — What is the overall no-show rate?
Formula	$(\text{Total No-Shows} \div \text{Total Appointments}) \times 100$
Result	48.46%
Interpretation	Nearly 1 in every 2 appointments is missed. This is the headline metric and baseline for the entire project. All interventions will be measured against this figure. The target should be to bring this below 35% through data-driven changes to booking policy, reminders, and patient engagement.

KPI 2 — Reminder Effectiveness Rate

Detail	Description
Linked to	BQ2 — Do reminders reduce no-show rates and does channel matter?
Formula	No-Show Rate (No Reminder) – No-Show Rate (With Reminder)
Result	4.03 percentage point reduction overall SMS: –5.64 pp Email: –2.98 pp WhatsApp: –1.62 pp
Interpretation	Reminders work — but channel choice matters significantly. SMS produces the largest reduction and should be the default channel. WhatsApp underperforms relative to its usage volume (1,101 reminders sent), making it a poor return on effort. The clinic should shift its default channel to SMS.

KPI 3 — No-Show Rate by Booking Lead Time Band

Detail	Description
Linked to	BQ3 — Does booking lead time affect the likelihood of a no-show?
Formula	No-show rate calculated per booking lead time band (0–7, 8–14, 15–30, 31–60 days)
Result	0–7 days: 27.81% 8–14 days: 33.55% 15–30 days: 43.21% 31–60 days: 60.49%
Interpretation	This is the most actionable finding in the dataset. No-show rates more than double from same-week bookings to 5-to-8-week bookings. The clinic should strongly consider capping booking windows at 14 days for non-urgent appointment types, particularly Follow-ups. For appointments that must be booked further ahead, reminders should be timed closer to the appointment date — not just at the time of booking.

KPI 4 — No-Show Rate by Previous No-Show History

Detail	Description
Linked to	BQ4 — Do patients with prior no-shows have higher future rates?
Formula	No-show rate segmented by previous_no_shows count (0, 1, 2, 3+)
Result	Low Risk (0 prior): 43.51% Medium Risk (1 prior): 53.49% High Risk (2+ prior): ~60–100%
Interpretation	Prior no-show history is the strongest predictor of future no-show behaviour. Even one prior no-show increases the risk by nearly 10 percentage points. The clinic should flag Medium and High Risk patients for additional outreach before their next appointment — for example, a personal phone call in addition to the standard reminder.

KPI 5 — No-Show Rate by Appointment Type

Detail	Description
Linked to	BQ1 — How does no-show rate vary by appointment type?
Formula	No-show rate calculated per appointment_type category
Result	Follow-up: 51.23% Diagnostic Test: 49.75% Specialist: 47.44% General: 46.64%
Interpretation	Follow-up appointments have the highest no-show rate at 51.23%, well above the overall average. These are often the appointments most important for ongoing care, yet patients assign them the least urgency. A dedicated reminder strategy — including shorter booking windows and a message explaining why the follow-up matters — should be introduced specifically for this appointment type.

4.1 KPI Summary

KPI	Name	Result	Linked BQ	Priority
KPI 1	Overall No-Show Rate	48.46%	BQ1	Baseline
KPI 2	Reminder Effectiveness Rate	SMS: -5.64 pp WhatsApp: -1.62 pp	BQ2	High
KPI 3	No-Show Rate by Booking Lead Time	27.81% → 60.49% across bands	BQ3	Critical
KPI 4	No-Show Rate by Prior History	Low: 43.51% Medium: 53.49% High: ~63%	BQ4	Critical
KPI 5	No-Show Rate by Appointment Type	Follow-up: 51.23% General: 46.64%	BQ1	High

Part 5 — Business Insights and Recommendations

Five meaningful insights were identified from the analysis. Each insight includes the finding from the data, its implication for the clinic, and a concrete recommendation.

Insight 1 — The No-Show Problem is Critically Large and Requires an Active Strategy

Finding: 48.46% of all appointments result in a no-show. Nearly 1 in every 2 scheduled slots is wasted.

Implication: This is not a minor operational inconvenience. At this scale, the clinic is effectively running at half capacity for a significant portion of its appointments. Revenue, staffing, and patient outcomes are all affected.

Recommendation: Set a formal no-show reduction target (e.g. reduce from 48.46% to below 35% within 6 months). Use KPI 1 as a monthly tracking metric. Introduce a dedicated no-show reduction programme combining booking policy, reminders, and patient risk flagging.

Insight 2 — Booking Lead Time is the Most Actionable Predictor

Finding: Same-week appointments have a 27.81% no-show rate. Appointments booked 31–60 days ahead have a 60.49% rate — more than double.

Implication: The further in advance an appointment is booked, the more likely it is to be forgotten, deprioritised, or conflicted with something else. This pattern is clear, consistent, and directly within the clinic's control.

Recommendation: Limit booking windows to a maximum of 14 days for Follow-up and General Consultation appointment types where operationally possible. For appointments that must be booked further ahead, send reminders 2–3 days before the appointment — not only at the time of booking.

Insight 3 — Patient Risk Can Be Identified Before the Appointment

Finding: Patients with 1 prior no-show: 53.49% rate. Patients with 2+ prior no-shows: ~60–100% rate. Patients with no prior no-shows: 43.51%.

Implication: The clinic already holds the information needed to identify high-risk patients before each appointment. This information is sitting unused in the dataset.

Recommendation: Implement a simple patient risk tier flag (Low / Medium / High) in the clinic booking system. For Medium and High Risk patients, add a personal phone call or a second reminder message 24 hours before the appointment in addition to the standard reminder.

Insight 4 — SMS Reminders Are the Most Effective Channel and Should Be the Default

Finding: SMS: 45.75% no-show rate (–5.64 pp vs no reminder). Email: 48.41% (–2.98 pp). WhatsApp: 49.77% (–1.62 pp).

Implication: WhatsApp reminders — which account for 1,101 of the 3,634 reminders sent — produce almost no reduction compared to sending no reminder at all. This represents a significant waste of effort.

Recommendation: Switch the default reminder channel from WhatsApp to SMS for all patients who have a mobile number on record. Review the reminder system configuration and reallocate WhatsApp reminders to SMS where possible. Track channel effectiveness monthly using KPI 2.

Insight 5 — Follow-Up Appointments Need a Dedicated Engagement Strategy

Finding: Follow-up appointments have the highest no-show rate at 51.23% — above the overall average of 48.46% and the highest of all four appointment types.

Implication: Follow-ups are often the most clinically important appointments for patient continuity of care, yet they are the most likely to be skipped. Patients may not understand the importance of returning for a follow-up after an initial consultation.

Recommendation: Create a dedicated follow-up appointment reminder that explains why the follow-up is important for the patient's health — not a generic reminder. Combine this with shorter booking windows (maximum 7 days where possible) to reduce the forgetting effect. Consider an opt-in confirmation call for all follow-up appointments.

Part 6 — Cross-Track Collaboration

The Week 5 assignment requires each intern to identify and document at least one meaningful collaboration point with another project track. This simulates a real multidisciplinary project environment.

Item	Detail
Track collaborated with	Data Science Track
What was shared	KPI results and variable importance findings — specifically that <code>booking_lead_days</code> , <code>previous_no_shows</code> , and <code>reminder_sent</code> are the strongest predictors of no-show behaviour in the dataset.
Why it was relevant	The Data Science track is building a no-show prediction model. The KPIs and variable analysis from this track provide evidence for which features to prioritise in the model. Without this analytical input, feature selection would be guided by assumption rather than actual data evidence.
What changed as a result	The <code>lead_time_band</code> and <code>patient_risk_tier</code> derived columns created in the Data Analytics notebook are directly reusable as engineered features for the Data Science prediction model. These are structured, categorical variables already aligned to the no-show target — reducing the Data Science team's feature engineering workload.

Part 7 — Assumptions, Limitations, Risks and Dependencies

This section reviews every item identified in Week 4 and documents what has changed, been resolved, or newly emerged during Week 5.

Item	Status	Week 5 Update
Dataset accurately represents appointment patterns	Still relevant	Accepted for this analysis — no evidence to the contrary
appointment_outcome is correctly recorded	Still relevant	No evidence of misclassification found during EDA
reminder_channel nulls explained by reminder_sent = No	Resolved	Confirmed in data preparation — structurally valid
Cancelled treated separately from No-Shows	Still relevant	Cancellations excluded from all KPI calculations and analysed separately in EDA
Dataset is fictional — findings are illustrative only	Still relevant	All findings framed as illustrative throughout the report
No-show reasons are unknown	Still relevant	Not captured in the dataset — a permanent limitation
90 missing distance values	Resolved	Filled with column mean (10.11 km) during data preparation
Date format errors risk	Resolved	Both date columns converted to datetime in data preparation
Cancelled records distorting no-show rate	Resolved	Cancelled excluded from all no-show rate calculations
NEW: WhatsApp reminder channel underperforms	New finding	WhatsApp (49.77%) barely improves on no reminder (51.39%) — needs channel strategy review
NEW: Lead time band pattern is extremely strong	New finding	No-show rate doubles from 0–7 to 31–60 day band — stronger than anticipated in Week 4

Part 8 — Week 5 Project Summary

8.1 What Was Planned for Week 5

Complete data preparation, run full exploratory data analysis across all key dimensions, calculate all 5 KPIs with interpretations, produce visualisations, and document 5 business insights and recommendations.

8.2 What Was Actually Completed

- Data preparation — date columns converted, missing values handled, two derived columns created (lead_time_band, patient_risk_tier).
- Full EDA across 12 dimensions — appointment type, day, time, age group, reminders, channel, lead time, prior history, risk tier, distance, and cancellations.
- All 5 KPIs calculated and interpreted with supporting data.
- 8 charts produced and saved.
- 5 business insights and recommendations documented.
- Cross-track collaboration with Data Science track completed and documented.

8.3 Key Findings

- 48.46% overall no-show rate — nearly 1 in 2 appointments missed.
- Booking lead time is the most actionable predictor: same-week bookings have a 27.81% no-show rate vs 60.49% for bookings made 31–60 days ahead.
- Prior no-show history reliably identifies high-risk patients — rate rises from 43.51% (no prior NS) to 100% (5 prior NS).
- SMS is the most effective reminder channel (45.75% NS rate, –5.64 pp vs no reminder).
- WhatsApp reminders significantly underperform — barely better than sending no reminder.
- Follow-up appointments have the highest no-show rate (51.23%).

8.4 Major Challenges Encountered

- Date columns required conversion before any time-based analysis could run.
- Deciding how to handle Cancelled appointments — resolved by treating them entirely separately from No-Shows.
- Selecting meaningful cut-off points for booking lead time bands — resolved by using natural booking windows (same week, fortnight, month, two months).

8.5 Important Decisions Made and Why

Cancelled excluded from no-show KPIs: A cancellation is not the same as a no-show — the patient gave notice. Including them would inflate the no-show rate and misrepresent the problem.

Missing distance values filled with column mean: Preferred over dropping 90 rows to preserve sample size. The gap is small (1.8%) and the mean is a reasonable estimate.

Risk tier classification created: Grouping patients into Low / Medium / High risk makes the prior no-show finding immediately usable by clinic staff, not just analysts.

8.6 Changes to the Week 4 Approach

No major changes to the planned phased approach. One unplanned finding emerged: WhatsApp reminders significantly underperform relative to their volume, which was not anticipated in Week 4. This became a dedicated recommendation in Week 5.

8.7 Cross-Track Collaboration Completed

KPI results and variable importance findings were shared with the Data Science track. The `lead_time_band` and `patient_risk_tier` derived columns are directly reusable as engineered features for the no-show prediction model.

8.8 Remaining Work

- Power BI dashboard development — planned for Week 7.
- Deeper time-based trend analysis using the converted date columns.
- Month-over-month no-show rate trends.
- Detailed reminder channel analysis broken down by appointment type.

8.9 Proposed Focus for Week 6

- Begin Power BI data model and dashboard structure.
- Deepen reminder channel analysis by appointment type and age group.
- Explore time-based trends using `appointment_date` and `booking_date`.
- Document findings in Week 6 project summary report.